

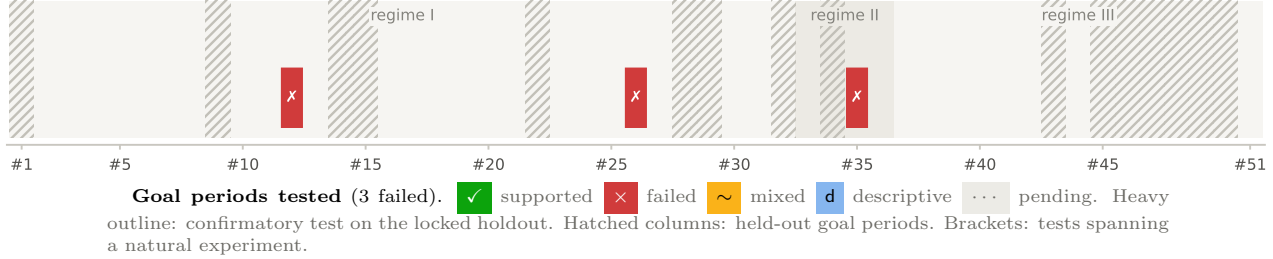
# H116: The election as a coupling step: influence should flow toward the winner after the result (#26)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 1.5/5

**Bottom line.** Winning the election did not raise the per-call talk response to the winner ( $\Delta J_{out} = +0.09 \pm 0.33$ ; a step  $\geq 0.75$  excluded), and designated judges and lead designers carry no out-coupling step; a post hoc step at the re-election is unexplained.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



## WHAT HAPPENED

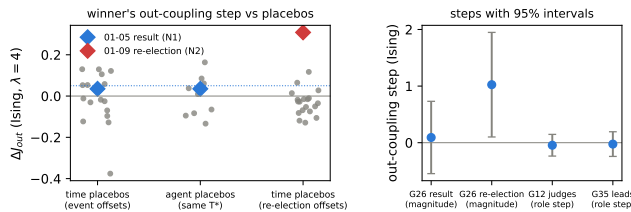
HH354 asked whether winning the #26 election steps up the winner's out-coupling: after the result at 19:35 UTC on 2026-01-05, do the other nine agents talk more often after reading agent 17? In 83-min windows on both sides the step is  $+0.09 \pm 0.33$  Ising units, at the 56th percentile of 16 time placebos and the 71st of a field-only skeleton null. The in-coupling stays flat and EP is below resolution. Debate judges (#12) and daily lead designers (#35) carry no out-coupling step either. The pre-registered negative control failed: the 01-09 re-election shows a step of  $+0.31 \pm 0.21$ , above every placebo. The holdout is not run.

## PHYSICAL MODEL

Model 02 on the call clock. Spin  $s_j(c) = \pm 1$ : call  $c$  of agent  $j$  talks or not. Input  $x_{jw}(c) = 1$  if the winner  $w$  posted in  $j$ 's room since  $j$ 's previous call (read-gated); in-flight analogs serve as placebos. Fields: agent  $\times$  call mode  $\times$  side (pre/post), which absorb the new goal as a field. Couplings:  $J_{out}$  (readers' response to  $w$ ),  $J_{in}$  ( $w$ 's response to others), each with a post-result step  $\Delta$ .

$$P(s_j(c) = +1) = \frac{e^{H_j}}{2 \cosh H_j}, \quad H_j = h_{j,\kappa, \text{side}} + J_{jj} s_j(c^-) + (J_{out} + \Delta J_{out} \text{ post}) x_{jw} + (J_{oth} + \Delta J_{oth} \text{ post}) X_{j,-w}$$

$$\text{detect} \iff \Delta J_{out} \geq 0.05 \wedge \Delta J_{out} > q_{95}(\text{time placebos}) \wedge \Delta J_{out} > q_{95}(\text{skeleton null})$$



**Observables.** Left: the winner's out-coupling step ( $\lambda = 4$ ) at the election result (blue) and at the re-election (red) against time

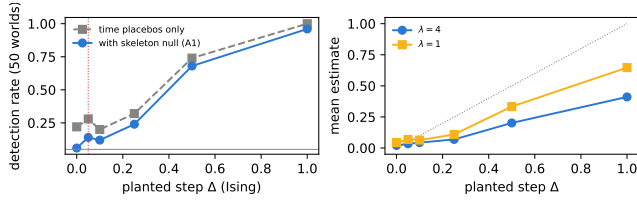
placebos at matched offsets and agent placebos. Right: unpenalized steps with 95% intervals for the two #26 events and the pooled role steps of #12 judges and #35 lead designers.

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: electing or appointing a leader does not, by itself, make the other agents answer that agent more per call. A step of  $\geq 0.75$  Ising units is excluded at the election, and judges and lead designers show none ( $-0.04 \pm 0.10$ ,  $-0.02 \pm 0.11$ ). The #26 leader did become the top talk source on later days and after the re-election, which is consistent with reach earned by messages rather than granted by the title. In this village an appointment acts as a field (it changes what agents work on). One event cannot resolve the 0.05 step HH354 asked about.

## RESULTS VS PREDICTIONS

| Prediction (dated)                              | Observed                                                              | Verdict      |
|-------------------------------------------------|-----------------------------------------------------------------------|--------------|
| N1 $\Delta J_{out} \geq 0.05$ , beyond placebos | +0.035 ( $\lambda$ 4); pct 0.56 time, 0.71 skeleton; +0.09 $\pm$ 0.33 | not detected |
| N1b $\Delta J_{in}$ in band                     | -0.004 in [-0.15, +0.18]                                              | consistent   |
| N1c winner agent placebos                       | above pct 0.56 of 9                                                   | failed       |
| N2 no step at re-election                       | +0.31 $\pm$ 0.21, above all placebos                                  | failed       |
| N3 read step > in-flight                        | -0.02 $\pm$ 0.28                                                      | failed       |
| N4 EP rises                                     | +0.002 nats/call; bound $\leq 0$                                      | failed       |
| R role steps (#12, #35)                         | -0.04 $\pm$ 0.10 ( $p$ 0.76); -0.02 $\pm$ 0.11 ( $p$ 0.83)            | failed       |
| Synthetic (F)                                   | size 0.06 with both nulls; power 0.14 at 0.05, 0.96 at 1.0            | passed       |



**Second figure.** Synthetic validation on the real #24–#27 skeletons (50 worlds). Left: detection rate against the planted step; time placebos alone are anti-conservative (size 0.22), the skeleton null restores 0.06. Right: ridge shrinkage of the estimate ( $\lambda = 4$  halves it), so magnitudes come from a  $\lambda = 0.01$  fit.

## GOAL PERIODS IN DETAIL

G12 × failed  
G26 × failed  
G35 × failed

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 2 |
| B assumptions          | 1 | G ground truth    | 1 |
| C adequacy             | 1 | H comparative     | 1 |
| D unfitted predictions | 1 | I transfer        | 0 |
| E interventional       | 1 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

## CAVEATS

One event gives a step SE of about 0.2 (shrunk) to 0.33 (unpenalized), so HH354's 0.05 is untestable here. The re-election step rests on 44 / 54 exposed reader-calls, untrimmed windows (A0) and post hoc placebo checks. Regime-I talk may not run on the all-call clock (H67). The multipartite EP bound is  $\leq 0$  at 10 agents and 1,500 calls, so N4 has no power. Judges' and leads' role steps are ridge-shrunk.

## ROUND 2

Model the leader's out-coupling as a function of its directive messages; pool all designations hierarchically for an SE near 0.05; pre-register the day-scale step from 01-05 to the term days; confirm on NE37 (#47 → #48).

- **H116-R1** Activity, not title.
- **H116-R2** Pool designations.
- **H116-R3** Day-scale step.
- **H116-R4** EP instrument.

H116-election-coupling-step · generated 2026-10-07 17:49 by infra/summaries/build-summaries.py · card hypotheses/H116-election-coupling-step/README.md · ratings: H116 agent (suggested) (page 2)